Project folder anatomy

A SWAT+ project is a single flat folder of plain-text files. The executable reads everything in that folder and writes outputs back into the same folder.

There are no subdirectories. The grouping below is by purpose, not by location.

The master file: `file.cio`

`file.cio` is the index. It tells SWAT+ which file names to read for each category of input. Every line names a category and the file(s) to read. The literal token `null` means "no file for this slot". Full column reference: file.cio.

Excerpt from `Ames_sub1/file.cio`:

```
file.cio: AMES
simulation        time.sim          print.prt         null              object.cnt        null
basin             codes.bsn         parameters.bsn
climate           weather-sta.cli   weather-wgn.cli   null              pcp.cli           tmp.cli
connect           hru.con           null              null              null              null
hru               hru-data.hru      null
aquifer           nbull             null
hydrology         hydrology.hyd     topography.hyd    null
hru_parm_db       plants.plt        fertilizer.frt    tillage.til       pesticide.pes     null
ops               harv.ops          graze.ops         irr.ops           chem_app.ops      fire.ops
lum               landuse.lum       management.sch    cntable.lum       cons_practice.lum ovn_table.lum
init              plant.ini         soil_plant.ini    null              null              null
soils             soils.sol         nutrients.sol     null              null
decision_table    lum.dtl           null              null              null
regions           ls_unit.ele       null              null              null              null
```

Read this file first when you open a project. It tells you what every other file is for.

File groups

Simulation control

File	What it controls
`time.sim`	Start and end dates, time step
`print.prt`	Which output files to write, and at which aggregation (`_day`, `_mon`, `_yr`, `_aa`)
`object.cnt`	Object counts (how many HRUs, channels, aquifers, etc.)

Basin parameters

File	What it controls
`codes.bsn`	Method switches (which equation set to use for ET, runoff, routing, …)
`parameters.bsn`	Basin-wide numeric parameters

Climate

File	What it controls
`weather-sta.cli`	List of weather stations and which files belong to each
`weather-wgn.cli`	Weather generator parameters (used to fill missing data)
`pcp.cli`, `tmp.cli`, …	Indexes pointing to the actual `.pcp`, `.tmp`, etc. data files
`<name>.pcp`, `<name>.tmp`	The daily precipitation and temperature time series

Spatial objects and connectivity

File	What it controls
`hru.con`	HRU connectivity (which object each HRU drains to)
`chandeg.con`, `aquifer.con`, `reservoir.con`	Channel, aquifer, reservoir connectivity
`rout_unit.con`	Routing unit connectivity
`hru-data.hru`	Per-HRU pointers to topography, soils, land use, management

Hydrology

File	What it controls
`hydrology.hyd`	Per-HRU hydrologic parameters
`topography.hyd`	Per-HRU slope, slope length, field length

Soils

File	What it controls
`soils.sol`	Soil layer properties (bulk density, available water, conductivity, …)
`nutrients.sol`	Initial soil nutrient pools

Land use and management

File	What it controls
`landuse.lum`	List of land use scenarios and the files that define each
`management.sch`	Operation schedules (planting, harvest, fertilizer, …)
`cntable.lum`	Curve number lookup
`cons_practice.lum`	Conservation practice factor
`ovn_table.lum`	Manning's n for overland flow
`lum.dtl`	Decision tables that trigger operations based on conditions

Plant, fertiliser, tillage, pesticide databases

File	What it controls
`plants.plt`	Plant parameter database
`fertilizer.frt`	Fertiliser composition
`tillage.til`	Tillage operation effects
`pesticide.pes`	Pesticide properties

Initial conditions

File	What it controls
`plant.ini`	Starting plant communities by HRU
`soil_plant.ini`	Starting soil water, residue, plant biomass

Aquifers, reservoirs, channels

File	What it controls
`aquifer.aqu`	Aquifer parameters
`reservoir.res`, `initial.res`	Reservoir parameters and initial state
`channel-lte.cha`, `hyd-sed-lte.cha`	Legacy channel routing
`hydrology.cha`, `sediment.cha`, `nutrients.cha`	SWAT+ channel routing parameters

Outputs (written by the model)

Files ending in `_day.txt`, `_mon.txt`, `_yr.txt`, or `_aa.txt`. The prefix tells you the spatial scope (`basin_`, `region_`, `lsunit_`, `hru_`, `channel_`, `aquifer_`, `reservoir_`, …) and the underscore code tells you the variable group:

Code	Variables
`_wb`	water balance
`_nb`	nutrient balance
`_ls`	landscape (sediment)
`_pw`	plant and water stress
`_aqu`	aquifer
`_cha` / `_sd_cha`	channel
`_res`	reservoir

See Output reference for column details.

Diagnostics

File	What it tells you
`simulation.out`	Run progress and successful completion message
`diagnostics.out`	Warnings and unexpected values during the run
`checker.out`	Validation messages produced before the simulation starts
`area_calc.out`, `files_out.out`, `erosion.txt`	Setup summaries

Always check these three (`simulation.out`, `diagnostics.out`, `checker.out`) when a run does not behave as expected.

Next

The full per-file reference lives in Input reference, with one page per file and a real example pulled from `Ames_sub1`.